Yingrui Hou

Data Science & Scientific Data Engineering

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https://yingrui-hou.github.io/PersonalPage/

Profile

Physics PhD with postdoctoral experience in reproducible data analysis, model validation and performance diagnosis. Built CatBoost workflows and calibrated learned models. Starting with no TensorFlow knowledge, adapted research code, added models, verified it with simulations and migrated it to a shared GPU workstation. Brings analysis coordination and international communication to data science and scientific data engineering.

Core skills

- Model assessment: deep-learning calibration, image-recognition model validation, real-data performance checks, train/test diagnostics and bias correction.
- Modelling: likelihood and mixture models, statistical inference, uncertainty analysis, CatBoost/gradient boosting, feature selection and distribution reweighting.
- Workflows: inherited-code adaptation, reusable processing, Shell/YAML/CMake, Linux and shared GPU workstation setup/management.
- Programming & computing: Python, NumPy, pandas, scikit-learn, statsmodels; C/C++, ROOT/RooFit; TensorFlow and GPU numerical evaluation.

Research experience

Postdoctoral Researcher | 2021–2026

CNRS — LAPP, Annecy (2021–2023); LPC/LPCA, Clermont-Ferrand (2023.12.01–2026.05.31). Research in LHCb and GRAiNITA collaborations.

- Developed a Shell pipeline for beam-test acquisition, storage, quick calibration and initial quality checks, connecting detector readout, tracking and timing calibration. Participated in on-site setup and data taking; assisted subsequent fitting and measurement-effect corrections.
- Trained CatBoost with pre-training reweighting, correlation checks, five-fold feature selection and subsequent hyperparameter tuning; set the threshold using fitted event yields and selection efficiencies.
- Aligned simulated, control and target datasets through gradient-boosting reweighting and calibration; propagated residual modelling differences into uncertainty assessments.
- Diagnosed incorrect ECAL photon truth associations; revised cluster-energy matching, tested multiple simulated decay channels and improved thread safety through review. The merged change supported subsequent Run 3 analyses and reconstruction-performance checks.

Doctoral research training | 2016–2021

University of Chinese Academy of Sciences — joint Master and PhD in Particle Physics. Training in quantitative problem formulation, scientific computation, experimental validation and communication of research results.

Education

- Joint Master and PhD in Particle Physics — University of Chinese Academy of Sciences, 2016–2021.
- BSc in Applied Physics — China University of Mining and Technology, 2010–2014.

Selected project evidence

ECAL diagnosis and integrated software correction

Traced anomalous simulated photon resolution to incorrect truth associations under high occupancy. Revised matching using normalised truth-particle energy deposits within the photon cluster. Checked identities, energies and event histories across several simulated decay channels. Improved thread safety following review; the MR was merged and used in Run 3 analyses and reconstruction-performance checks.

Calibrating learned-model outputs

Validated and calibrated a deep-learning flavour-tagging model using real data for subsequent analysis. Transferable method: check the reliability of learned-model outputs before using them in a new analysis.

Machine learning under dataset shift

Selected Bs → phi gamma with simulated signal and real-data background. Checked correlations and sample shifts using control data and hep_ml reweighting. Used five folds for feature selection and subsequent ROC/AUC tuning. Scanned a precision-oriented threshold using fitted baseline yields and efficiencies; the final application sample was excluded from classifier training and hyperparameter tuning.

Learning TensorFlow and delivering GPU computation

Learned TensorFlow from inherited code; personally ported amplitude models, added efficiency, resolution and bias effects, and validated calculations against mature C++ code at identical sampling points. Resolved discrepancies by aligning variable types; checked simulated observables against real data with KS tests. Set up a shared GPU workstation in half a day, then achieved a GPU run in another half-day. Contributed to testing a collaborator-developed fitter.

For flavour tagging and ECAL clustering, participated in model-construction and training discussions for downstream calibration and validation; did not train these two models.

Learning, delivery & collaboration

- Learning by implementation: extended unfamiliar TensorFlow code with new research models and checked it on generated data; applied CatBoost and Geant4 in other projects.
- Delivery: agreed component interfaces and variable definitions before coding; reassessed priorities and timelines with collaborators as data processing exposed new issues.
- Collaboration: GitLab development and mutual merge-request review; identified a fitter sign error through mathematical checks. Weekly planning and rapid technical discussions; research presentations at GDR-InF 2022 and FCC/DRD France 2025.

Proposed first assignment: agree the decision and success criteria, check data quality, establish a baseline and deliver a focused validation study with findings and next steps.

Languages & mobility

Chinese: native | English: C1 | French: B1–B2. Long-distance cycling and travel; open to relocation in Europe.

Project references

- ECAL diagnosis and integrated software correction (https://yingrui-hou.github.io/PersonalPage/index.html#work-6)
- Calibrating learned-model outputs (https://yingrui-hou.github.io/PersonalPage/index.html#work-9)
- Machine learning under dataset shift (https://yingrui-hou.github.io/PersonalPage/index.html#work-3)
- Learning TensorFlow and delivering GPU computation (https://yingrui-hou.github.io/PersonalPage/index.html#work-8)